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Valuing Snap After the IPO Quiet Period (A)

On March 2, 2017, Snap Inc., the disappearing message app, went public at \$17.00 per share on the New York Stock Exchange (NYSE). According to *The Wall Street Journal*, “investors clamored for a piece of the fast-growing social network,” causing the price to surge and eventually close at \$24.48, up 44% on the day.¹ At that price, the firm had a market capitalization of \$34 billion and its two 20-something founders, Bobby Murphy and Evan Spiegel, became the youngest self-made billionaires in the country to own a public company.² Over the next three weeks, Snap traded as low as \$19 and as high as \$27, and closed at \$22.74 on Friday, March 24. During this time, 16 analysts made investment recommendations on Snap: two with buy recommendations, seven with holds, and seven with sells.

Analysts at firms that served as underwriters of Snap’s initial public offering (IPO), including Morgan Stanley and Goldman Sachs (the lead underwriters), were forbidden from issuing reports on Snap until 25 days after the IPO according to Securities and Exchange Commission (SEC) rules.^a When the “IPO quiet period” ended on Monday, March 27, 14 more firms issued reports with recommendations—ten with buy recommendations and four with holds—with price targets ranging from \$21 to \$31. Brian Nowak, Morgan Stanley’s internet analyst, had a price target of \$28 and an overweight recommendation. He said, “We are bullish on Snap’s ability to monetize its highly engaged daily active user (DAU) base.”³

As she sat at her desk, Elizabeth Kemp scanned the reports searching for insights. Kemp was the portfolio manager of a \$400 million long-only, technology fund at Sand Hill Road Capital, and had bought 500,000 Snap shares at the IPO. With a gain of almost \$3 million ($= (\$22.74 - \$17.00) \times 500\text{K}$ shares), she was inclined to buy more. But she recalled seeing a report written on March 14 by Kip Paulson, Cantor Fitzgerald’s internet analyst with a price target of \$18.00 and an underweight (sell) recommendation based on concerns about Snap’s unproven business model, untested management team, slowing growth, and fierce competition from larger rivals like Facebook/Instagram and Twitter.⁴

^a Although the SEC and FINRA had reduced the quiet period from 25 days to 10 days in 2015, underwriters still followed the 25-day rule because of prospectus delivery regulations (i.e., the prospectus had to be available to investors as the only official source of company information for 25 days after the IPO). See “Equity Research,” FINRA Regulatory Notice 15-20, August 2015, pp. 5-6. Available at: http://www.finra.org/sites/default/files/notice_doc_file_ref/Regulatory-Notice-15-30.pdf.

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As a relatively new portfolio manager with a good three-year track record, Kemp wondered if now was the time to harvest her gain or double down and buy more shares.

Company Background

Evan Spiegel met Bobby Murphy and Reggie Brown when they were undergraduate students at Stanford University.⁵ They created a disappearing picture app in 2011 called Picaboo with Spiegel serving as the chief executive officer, Murphy as the chief technology officer, and Brown as the chief marketing officer.⁶ Soon thereafter, Brown left the company, the founders changed the name to Snapchat, and they moved the operations to Los Angeles. Their app enabled users to send self-deleting images, called “snaps,” to other users. Within a few years, users could also send videos and could retain postings using the “my story” feature (see **Exhibit 1** for a description of the app).⁷

In 2016, the founders once again changed the name, this time to Snap Inc. The firm’s chief strategy officer, Imran Khan, explained why: “[W]e are bigger than just one app...[W]e believe that reinventing the camera represents our greatest opportunity to improve the way people live and communicate. Our products empower people to express themselves, live in the moment, learn about the world, and have fun together.”⁸ The founders described the strategy—a kind of virtuous cycle—this way: “Our strategy is to invest in product innovation and take risks to improve our camera platform. We do this in an effort to drive user engagement, which we can then monetize through advertising. We use the revenue we generate to fund future product innovation to grow our business.”⁹

The firm’s advertising revenue could be measured by the number of daily average users (DAU) times the average revenue per user (ARPU). At the end of 2016, it had advertising revenue of \$378 million based on 158 million DAU’s and an annual ARPU of \$2.39. Morgan Stanley’s Nowak predicted Snap would have revenue of \$4.9 billion by 2020, based on 236 million DAU’s and an ARPU of \$20.72. **Exhibit 2** shows Nowak’s revenue forecasts for 2016 and 2020. To illustrate Snap’s growth potential, Nowak compared Snap against three other social media companies—Instagram, Facebook, and Twitter (see **Exhibit 3**). Although Snap’s current ARPU was considerably less than its peers primarily due to its low “ad load” (advertisements per DAU per hour), Nowak predicted Snap would increase its ad load by more than 10 times over the next four years.

In 2016, Snap generated revenues of \$405 million and a loss of \$515 million. The good news, however, was that it had grown revenues by almost 600% while increasing losses by only 38% (see **Exhibit 4**). In fact, quarterly analysis showed growth in both DAUs and ARPU over the last three years (see **Exhibit 5**). To fund its growth, Snap raised \$1.8 billion of equity in 2016 from a group of venture capital firms.¹⁰ Having raised eight rounds of equity capital totaling \$2.6 billion,¹¹ Snap’s balance sheet was entirely debt-free (see **Exhibit 6**). Nevertheless, the finance team had lined up a \$1.1 billion unsecured revolving credit facility in July 2016, from a syndicate of lenders including Morgan Stanley, Goldman Sachs, Deutsche Bank, and Credit Suisse.¹² In terms of control, the firm had three classes of common stock, two of which had voting rights (Classes B and C) and one of which did not (Class A).

The IPO

In October 2016, Snap chose Morgan Stanley and Goldman Sachs to be the lead underwriters for an IPO scheduled for early 2017.¹³ Underwriters performed many roles: they conducted due diligence on the firm and its financial condition; helped write and file documents needed to issue public equity; created a syndicate of other brokerage firms to distribute the shares; and contacted prospective investors to gauge interest in the offering and to determine an offering price. Following the IPO, the underwriters often bought shares to stabilize the price and provided research coverage for clients.¹⁴

For these services, the Snap underwriters—a collection of 26 brokerage firms—would get a fee of 2.5% of the amount raised in the offering.¹⁵ The stated purpose of the offering was to increase the firm's capitalization and enhance its financial flexibility.¹⁶

In February 2017, Snap announced that it would sell 145 million new Class A non-voting common shares, and existing shareholders would sell another 55 million Class A shares.¹⁷ Spiegel and Murphy each planned to sell 16 million Class A shares.¹⁸ Nevertheless, they would still control 89% of the voting rights following the IPO because they owned the majority of the Class B and C shares.¹⁹ In total, there would be 1.4 billion (fully diluted) shares outstanding after the offering. While it was becoming more common to issue shares with limited voting rights, this was the first IPO on a US exchange of a stock with no voting rights.²⁰

Besides the absence of voting rights, analysts raised three other concerns about the offering. First, the price seemed high. Whereas Facebook, Twitter, Alphabet and Yelp traded at an average of 4.5 times 2018 revenues (enterprise value), Snap would trade at 13.9 times expected 2018 revenues according Paulson at Cantor Fitzgerald.²¹ Second, it had strong competitors—Facebook/Instagram, Google/YouTube, and Twitter—all vying for the same digital advertising dollars.²² And third, Snap relied on powerful suppliers: it had recently signed long-term contracts worth \$3 billion with Google Cloud and Amazon Web Services to provide computing, storage, bandwidth, and other services.²³

Despite these concerns, Snap closed up 44% at \$24.48 on its first day of trading. Because the offering was oversubscribed by 10 times,²⁴ the underwriters exercised their “greenshoe” or overallotment option. This contractual provision in the underwriting agreement gave them the ability to sell an additional 30 million Class A shares at the offering price and to collect additional underwriting fees.²⁵

The Analyst Quiet Period

During the next few weeks, “sell-side” analysts not involved with the underwriting process released reports, some of which were very critical of the firm and its prospects (see **Exhibit 7**).^b For example, Paulson wrote, “The (valuation) numbers do not add up... The company has much work ahead to prove that ROI [Return on Investment] is not only positive, but competitive with other platforms chasing these ad dollars.”²⁶ Another analyst warned that, “Snap is highly risky and not for those who want to sleep peacefully at night....”²⁷ Despite these concerns, Snap closed at \$22.74 on Friday, March 24.

When the IPO quiet period expired on Monday, March 27, Snap's underwriters issued a flurry of reports. **Exhibit 8** summarizes the recommendations and 12-month price targets for some of these firms; **Exhibit 9** summarizes valuation assumptions for the firms that valued Snap using discounted cash flow (DCF) analysis. Like many of his peers, Nowak used DCF analysis to value Snap. He said:

“We prefer DCF-based valuation (the standard methodology across our coverage universe) but think it is useful to look at [the] Snap valuation in multiple ways, to sanity check our assumptions around our DCF...[W]e also support our PT [price target] with 4 additional valuation methodologies [e.g., FCF and EBITDA multiples]...”²⁸

^b Whereas “sell-side” analysts worked at brokerage firms and provided clients with investment recommendations, “buy-side” analysts worked at investment firms and make buy and sell recommendations to portfolio managers. See “What is the difference between a buy-side and a sell-side analyst?” January 22, 2018, available at: <https://www.investopedia.com/ask/answers/difference-between-buy-side-analyst-and-sell-side-analyst/>, accessed 4-13-18.

To value Snap, Nowak forecast free cash flows for 10 years (including the terminal year), used a discount rate (weighted average cost of capital or WACC) of 9.7%, and used a perpetual (terminal value) growth rate of 3.5% (see **Exhibit 10**). As reference points, **Exhibit 11** shows the WACC's used by Nowak to value other internet and social media firms. Based on his analysis, Nowak assigned a price target of \$28 per share and issued an overweight recommendation.²⁹ **Exhibit 12** shows the distribution of stock recommendations for all firms covered by Morgan Stanley and for its investment banking clients. Nowak concluded his report by saying:

“Snap’s engaged/hard-to-reach millennial users and unique video offerings should attract significant ad dollars ...[I]nvestors may push back that [the] Snap valuation looks rich at this multiple [price target equal to 19 times 2018 revenue per share] relative to the peer group. However, we think this valuation is appropriate given the optionality for materially higher monetization if Snap is able to deliver on user growth and advertising innovation.”³⁰

Because investment bankers at Morgan Stanley had been underwriters for the Snap IPO, Nowak was required to include a disclosure statement on the first page of his report (see Note 2 in **Exhibit 10**). Later in the report, he disclosed the fact that Morgan Stanley had a “significant financial interest in the debt securities of...Snap Inc. [through the credit facility]”³¹ Disclosures regarding possible conflicts of interest had been mandated by the SEC following a \$1.4 billion settlement with 10 brokerage firms, including Morgan Stanley, in 2003.³² The settlement also required firms to separate investment banking (corporate advisory and capital raising) from equity research both physically and operationally (i.e., to create a proverbial “Chinese wall” between the two groups and to ensure analyst compensation was not tied to banking fees). The intent was to prevent people from exchanging non-public information and to ensure stock recommendations were not “tainted by efforts to obtain investment banking fees.”³³ Nowak’s disclosures were also bound by Morgan Stanley’s Code of Conduct which required employees to “put clients first” and to avoid conflicts of interest (see Note 3 in **Exhibit 10**).

Like Nowak, equity analysts at other firms were equally bullish on Snap. They said:

RBC Capital: “Snap has become an innovation leader—for both consumers and advertisers—in arguably the single fastest [growing] advertising medium today—Mobile...We believe that if it sustains its current level of innovation, it can sustain premium growth for a long time and scale to profitability.”³⁴

Jefferies: “We believe Snap has all the ingredients to build a robust advertising business. Snap has a large audience (158MM DAUs), deep engagement, and robust data about its users...[and] should scale its business without significant capex investment.”³⁵

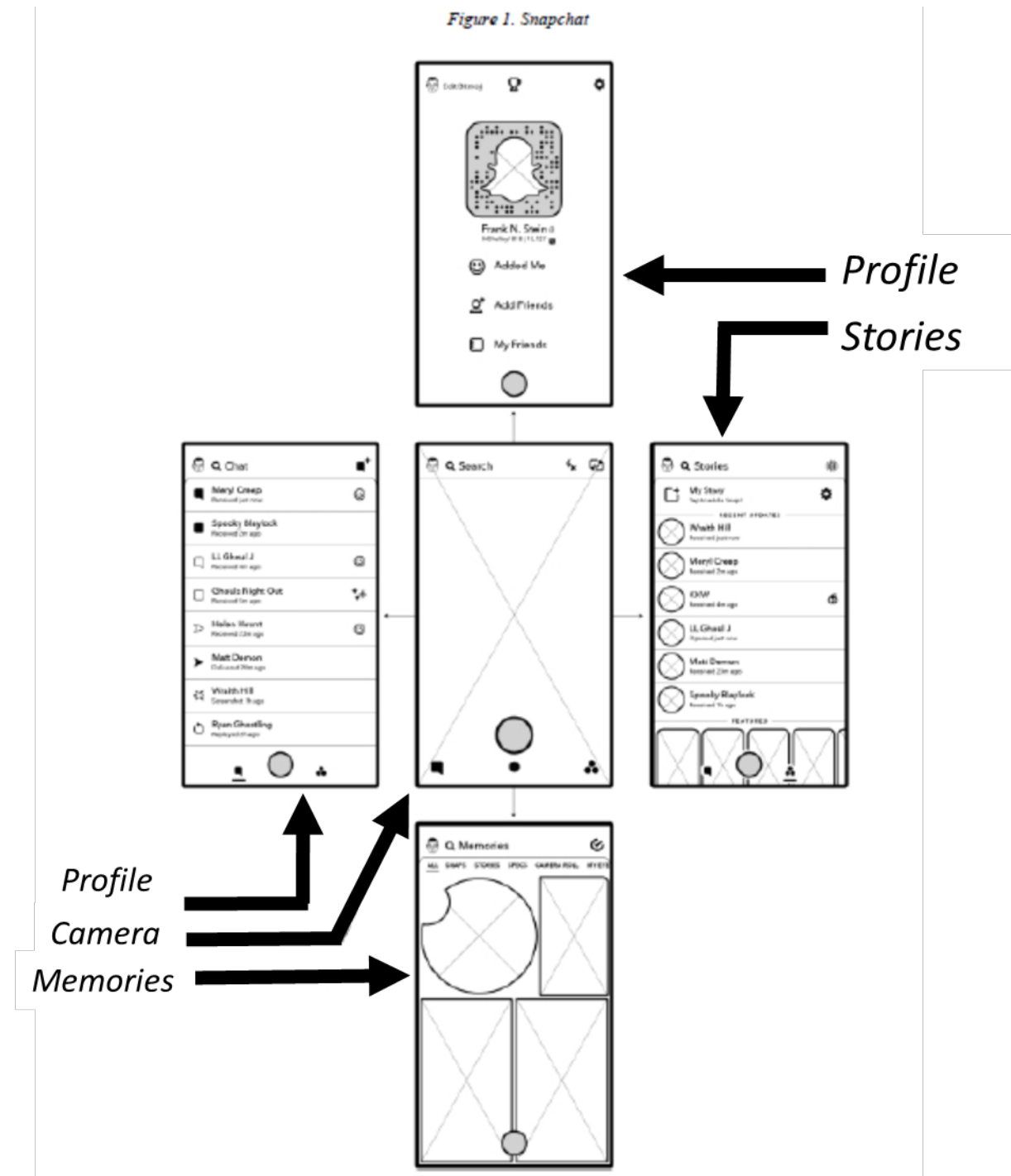
Deutsche Bank: “[We] see an attractive opportunity to buy the next great mobile ad business amidst a sea of investor skepticism...Competitive risks, while real, are overblown...”³⁶

Credit Suisse: “[W]e enthusiastically underwrite [Snap because] it is a scarce asset that offers advertisers access to a coveted younger demographic...”³⁷

Conclusion

Elizabeth Kemp sat at her desk wondering what to do. With so many new buy recommendations, Snap seemed poised for further price appreciation. Yet a few analysts remained skeptical. Having already made a good return on her investment, she wondered if now was the time to harvest her gains or to double down and buy more shares. As she pondered what to do, she recalled her experience with GoPro’s IPO in the summer of 2014: she bought shares at \$24 and it closed up 30% on the first day of

trading. After the quiet period expired, she sold all of her shares, in part because of the tepid ratings from analysts affiliated with the underwriters, only to see the price increase to \$80 within three months.³⁸ Along with this painful memory, she recalled a quotation attributed to Warren Buffett: “What we learn from history is that people don’t learn from history. When investors get too fearful or too greedy, they sometimes hide behind the notion that ‘this time is different.’ Usually they regret it.”³⁹

Exhibit 1 Snapchat App Overview

Source: Snap Inc., SEC Registration Statement, Form S-1, 2/2/17, p. 92.

Exhibit 2 Snap Advertising Revenue Model, 2016 Actual vs. 2020 Expected

Snap Advertising Revenue Model	2016	2020E	Notes
Daily Average Users (DAU, in millions)	158	236	User Base grows at 11% CAGR
X Hours per DAU per Day	0.42	0.50	
X Ads per DAU per Hour	0.6	7.9	Snap continues to grow ad load
X Average Pricing (\$ per 1K impressions)	\$26.00	\$14.37	Inventory scales and pricing falls
X 365 Days per Year	365	365	
Advertising Revenue (\$ mil)	\$378	\$4,889	Ad revenue grows at a 87% CAGR
Avg. Revenue per DAU (ARPU)	\$2.39	\$20.72	

Source: Nowak, B., et al., "Crackle or Pop? Initiate OW, \$28 PT," Snap Inc. analyst report Exhibit 6 (p. 7), Morgan Stanley Research, 3/27/18.

Exhibit 3 Snap Advertising Revenue Model vs. Comparable Internet Firms in 2016

2016 Advertising Revenue Drivers	Snap	Instagram	Facebook	Twitter
Daily Average Users (DAU, in millions)	158	345	1,227	129
X Hours per DAU per Day	0.42	0.36	0.58	0.15
X Ads per DAU per Hour	0.6	6.6	48.6	104.3
X Average Pricing (\$ per 1K impressions)	\$26	\$6	\$2	\$3
X 365 Days per Year	365	365	365	365
Advertising Revenue (\$ mil)	\$378	\$1,795	\$25,118	\$2,254
Avg. Revenue per DAU (ARPU)	\$2.39	\$5.20	\$20.47	\$17.47

Source: Casewriter estimates using Exhibit 5 (p. 7) in B. Nowak et al., "Crackle or Pop? Initiate OW, \$28 PT," Snap Inc. analyst report Exhibit 5 (p. 7), Morgan Stanley Research, 3/27/18.

Exhibit 4 Snap Inc. Income Statement (\$ millions)

	2015	2016
Revenue	\$58.7	\$404.5
Costs and Expenses		
Cost of Revenue	\$182.3	\$451.7
Research and Development	\$82.2	\$183.7
Sales and Marketing	\$27.2	\$124.4
General and Administrative	\$148.6	\$165.2
Total Costs and Expenses	\$440.4	\$924.9
Loss From Operations	(\$381.7)	(\$520.4)
Net Interest Income	\$1.4	\$3.3
Other Income (Expense), net	(\$0.2)	(\$4.6)
Loss Before Income Taxes	(\$380.5)	(\$521.7)
Income Tax Benefit (Expense)	\$7.6	\$7.1
Net loss	(\$372.9)	(\$514.6)

Source: Snap Inc. SEC Registration Statement, Form S-1, 2/2/17, p. 57.

Exhibit 5 Snap Inc. Quarterly Financial and Operating Statistics

	Quarterly Average (millions and dollars)					Financial Data (\$ millions)			
	Daily		Global		N. America				
	Active Users (DAU)	Growth in DAU	Avg. Rev. Per User (ARPU)	Growth in ARPU	Avg. Rev. Per User (ARPU)				
						Revenue	Cost of Revenue	Operating Loss	Gross Margin
4Q16	158	3%	\$1.05	25%	\$2.15	\$165.7	\$153.4	(\$169.7)	7.4%
3Q	153	7%	\$0.84	68%	\$1.76	\$128.2	\$127.8	(\$131.0)	0.3%
2Q	143	17%	\$0.50	56%	\$1.06	\$71.8	\$94.8	(\$115.9)	(32.0%)
1Q	122	14%	\$0.32	3%	\$0.67	\$38.8	\$75.8	(\$103.8)	(95.4%)
4Q15	107	14%	\$0.31	72%	\$0.65	\$32.7	\$64.8	(\$98.2)	(98.2%)
3Q	94	9%	\$0.18	200%	\$0.36	\$16.7	\$56.2	(\$96.9)	
2Q	86	8%	\$0.06	20%	\$0.13	\$5.3	\$40.6	(\$87.8)	
1Q	80	13%	\$0.05		\$0.10	\$3.9	\$20.7	(\$98.8)	
4Q14	71	15%							
3Q	62	9%							
2Q	57	24%							
1Q	46								

Source: Snap Inc. SEC Registration Statement, Form S-1, 2/2/17, pp. 62, 64, and 76.

Exhibit 6 Snap Inc. Balance Sheet (\$ millions)

	2015	2016
Assets		
Current Assets		
Cash & Securities	\$640.8	\$987.3
Accounts Rec.	\$44.3	\$162.7
Prepaid Expenses	\$7.4	\$30.0
Total Current Assets	\$692.6	\$1,180.0
Property and Equipment, net	\$44.1	\$100.6
Goodwill and Intangibles	\$177.1	\$395.1
Other Assets	\$25.1	\$47.1
Total Assets	\$938.9	\$1,722.8
Liabilities and Stockholders' Equity		
Current liabilities		
Accounts payable	\$0.7	\$8.4
Accrued exp. and other CL	\$155.6	\$148.3
Total Current Liabilities	\$156.3	\$156.7
Other liabilities	\$18.5	\$47.1
Total Liabilities	\$174.8	\$203.9
Stockholders' Equity	\$764.1	\$1,518.9
Total Liab. & Shareholders' Equity	\$938.9	\$1,722.8

Source: Snap Inc. SEC Registration Statement, Form S-1, 2/2/17, p. F-3.

Exhibit 7 Summary of Sell-Side Equity Analyst Ratings for Snap Inc. in March 2017

Date	Analyst Ratings in Existence on Each Date				Average Rating (Note a)	Average Price Target
	Buy	Hold	Sell	Total		
March 2	0	2	1	3	2.0	\$16.00
March 16	0	6	4	10	1.9	\$17.50
March 26	2	7	7	16	2.5	\$19.83
March 27	12	11	7	30	3.4	\$23.56

Source: Bloomberg, accessed 3/19/18.

Note (a): Numerical ratings: 1 = sell, 2 = underweight, 3 = neutral, 4 = overweight, and 5 = buy.

Exhibit 8 Summary of Equity Analyst Recommendations and Analysis of Snap Inc., March 2017

Analyst Firm & Role	Report Date	Under-Writer Share	Price Target	Analyst Recommendation
Independent Firms				
Morningstar Equity	1-Mar-17	none	\$15.00	n/a
Aegis Capital	2-Mar-17	none	\$22.00	Hold
Nomura Instinet	2-Mar-17	none	\$16.00	Reduce
Pivotal Research	2-Mar-17	none	\$10.00	Sell
Atlantic Equities	3-Mar-17	none	\$14.00	Underweight
Susquehanna	3-Mar-17	none	\$22.00	Neutral
FBN Securities	9-Mar-17	none	\$23.00	Perform
Cantor Fitzgerald	14-Mar-17	none	\$18.00	Underweight
Moffett Nathanson	16-Mar-17	none	\$15.00	Sell
Mizuho Securities	17-Mar-17	none	\$20.00	Neutral
Monness Crespi	20-Mar-17	none	\$25.00	Buy
Drexel Hamilton	21-Mar-17	none	\$30.00	Buy
Piper Jaffray	22-Mar-17	none	\$23.00	Neutral
Needham & Co.	22-Mar-17	none	n/a	Underperform
Summit Redstone	26-Mar-17	none	\$17.00	Sell
B of A/Merrill	n/a	none	n/a	Neutral
BMO Capital	n/a	none	n/a	Perform
Lead Underwriters				
Morgan Stanley	27-Mar-18	30.2%	\$28.00	Buy
Goldman Sachs	27-Mar-18	24.8%	\$27.00	Buy
Lead Managers (Book Runners)				
JP Morgan	27-Mar-17	13.3%	\$24.00	Neutral
Deutsche Bank	27-Mar-17	10.0%	\$30.00	Buy
Allen & Co.	<i>No research</i>	7.0%		n/a
Barclays	<i>No report</i>	6.0%		n/a
Credit Suisse	27-Mar-17	3.1%	\$30.00	Outperform
Other Underwriters (Co-Managers)				
Citibank	27-Mar-17	0.6%	\$27.00	Buy
Cowen & Co.	27-Mar-17	0.5%	\$26.00	Outperform
Jefferies	27-Mar-17	0.3%	\$30.00	Buy
JMP Securities	27-Mar-17	0.2%	\$28.00	Outperform
Oppenheimer	27-Mar-17	0.3%	\$21-\$26	Perform
RBC Capital	27-Mar-17	0.6%	\$31.00	Outperform
Stifel, Nicolaus	27-Mar-17	0.5%	\$24.00	Hold
UBS	27-Mar-17	0.6%	\$24.00	Neutral
William Blair	27-Mar-17	0.2%	\$25-\$28	Outperform

Source: Casewriter analysis of individual equity research reports on Snap Inc. from each firm, available on ThompsonOne, accessed 3/18/18. Certain data comes from public news articles about the IPO. Underwriter status was from: Balakrishnan, A., "Here's What Snap's Wall Street Underwriters Made Off the IPO," CNBC, 3/3/17, available at: <https://www.cnbc.com/2017/03/03/snap-ipo-what-wall-street-banks-made.html>, accessed 3/18/18.

Exhibit 9 Summary of Analyst Assumptions and Forecasts for Snap in 2020 (see Note 1 below)

Analyst(s) From	Report Date	Price Target	Discount	TV	Analyst Forecast for 2020			
			Rate WACC	Growth Rate	Revenue	DAU	ARPU	Free Cash Flow (FCF)
Independent Firms								
Pivotal	2-Mar-17	\$10	12.0%	6.0%	\$3,863	n/a	n/a	(\$178)
Aegis Capital	2-Mar-17	\$22	13.4%	3.0%	\$4,500	250	n/a	\$605
Susquehanna	3-Mar-17	\$22	n/a	n/a	n/a	280	\$21.24	\$357
FBN	9-Mar-17	\$23	10.3%	4.5%	\$4,816	369	\$14.09	(\$592)
Cantor	14-Mar-17	\$18	12.0%	4.0%	\$6,248	346	\$18.92	\$398
Piper Jaffray	22-Mar-17	\$23	10.7%	(2.0%)	\$6,396	289	\$22.41	\$253
Lead Underwriter								
Morgan Stanley	27-Mar-17	\$28	9.7%	3.5%	\$4,902	236	\$20.72	(\$243)
Lead Managers								
JP Morgan	27-Mar-17	\$24	13.0%	3.5%	\$6,808	299	\$21.30	\$1,026
Deutsche Bank	27-Mar-17	\$30	15.7%	n/a	\$5,126	272	\$18.06	\$1,275
Credit Suisse	27-Mar-17	\$30	11.0%	3.0%	\$4,634	281	\$17.02	\$351
Other Underwriters								
Cowen & Co	27-Mar-17	\$26	10.0%	n/a	\$5,204	289	\$18.16	\$351
JMP Securities	27-Mar-17	\$28	15.0%	n/a	\$4,965	n/a	n/a	\$634
Oppenheimer	27-Mar-17	range	n/a	n/a	\$5,649	281	\$18.89	\$953
RBC Capital	27-Mar-17	\$31	11.0%	5.0%	\$4,621	n/a	n/a	\$391
UBS	27-Mar-17	\$24	18.0%	n/a	\$5,269	332	\$16.70	\$442
William Blair	27-Mar-17	range	10.0%	n/a	\$4,654	n/a	n/a	n/a
Average		\$24.2	12.3%	3.4%	\$5,177	293.7	\$18.86	\$402
Median		\$24.0	11.5%	3.5%	\$4,965	285.1	\$18.89	\$391
Maximum		\$31.0	18.0%	6.0%	\$6,808	369.0	\$22.41	\$1,275
Minimum		\$10.0	9.7%	(2.0%)	\$3,863	236.0	\$14.09	(\$592)
Standard Deviation		\$5.5	2.5%	2.2%	\$795	38.3	\$2.44	\$487
Coeff. of Variation (note 2)		0.23	0.20	0.66	0.15	0.13	0.13	1.21

Source: Individual equity research reports on Snap Inc. from each firm, available on ThompsonOne, accessed 3/18/18.

Note 1: Includes only firms that described using discounted cash flow (DCF) analysis to value Snap. Not all firms revealed all of the information needed to conduct DCF analysis.

Note 2: The coefficient of variation is a unitless measure of dispersion. It is calculated by dividing the standard deviation of the sample by the mean.

Exhibit 10 Morgan Stanley's Cash Flow Forecasts and Valuation Assumptions for Snap Inc. (\$ millions)

Item	Actual	Financial Forecast (\$ millions)										TV Year (2026)
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025		
Income Statement												
Net Revenue	\$405	\$944	\$1,935	\$3,254	\$4,902	\$6,907	\$9,189	\$11,635	\$14,131	\$16,569	\$17,149	
Growth	589.5%	133.3%	105.0%	68.2%	50.6%	40.9%	33.0%	26.6%	21.4%	17.3%	3.5%	
Adj. EBITDA (Note 1)	(\$459)	(\$772)	(\$610)	(\$195)	\$363	\$1,474	\$2,616	\$3,888	\$5,223	\$6,574	\$6,804	
Adj. EBITDA Margin	(113.6%)	(81.8%)	(31.5%)	(6.0%)	7.4%	21.3%	28.5%	33.4%	37.0%	39.7%	39.7%	
Cash Flow Items												
- Cash Taxes	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$0	\$823	\$1,448		
- Incr. (Decr.) in Net WC	\$151	\$257	\$228	\$231	\$228	\$209	\$224	\$248	\$207	\$151		
- Capital Expenditures	\$67	\$83	\$97	\$109	\$120	\$130	\$140	\$148	\$155	\$161		
- Stock-based Comp.	\$32	\$75	\$159	\$203	\$257	\$317	\$421	\$533	\$648	\$759		
Free Cash Flow	(\$709)	(\$1,186)	(\$1,093)	(\$737)	(\$243)	\$818	\$1,831	\$2,958	\$3,390	\$4,054	\$4,196	
Morgan Stanley Valuation Assumptions												
Market Data (March 27, 2017)												
Cost of Debt	3.00%	# Fully Diluted Shares (millions)				1,404	US Treasury Yields			27-Mar		
Future Leverage (D/V)	2%	Excess Cash (\$ mil)				\$2,081	30-Year			3.06%		
Tax Rate	35%	Perpetual Growth Rate					10-Year			2.38%		
Cost of Equity	9.90%											
Equity Risk Premium	5.59%											
Stated WACC	9.7%											

Source: Nowak, B., et al., "Crackle or Pop? Initiate OW, \$28 PT," Snap Inc. analyst report Exhibit 13 (p. 11), Morgan Stanley Research, 3/27/17.

Note 1: Adjusted EBITDA is a non-GAAP item. It excludes stock-based compensation expense (a cash expense), but includes depreciation and amortization.

Note 2: **Morgan Stanley Equity Research Disclosure:** Morgan Stanley does and seeks to do business with companies covered in Morgan Stanley Research. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of Morgan Stanley Research. (p. 1 of the Snap Inc. analyst report, 3/27/17).

Note 3: **Morgan Stanley Code of Conduct:** Our values inform everything we do: *Putting Clients First, Leading with Exceptional Ideas, Doing the Right Thing and Giving Back*. Our Code of Conduct reflects our continued commitment to performing all our business activities in accordance with these core values and in full alignment with the letter and spirit of applicable laws, regulations and our policies.... *Putting Clients First* requires us to place our clients' interests first and avoid conflicts between their interests and ours. Available at: <https://www.morganstanley.com/about-us-governance/code-of-conduct>, accessed 3/18/18.

Exhibit 11 Assumptions Used by Morgan Stanley for Internet Stocks and Other Market Data

Company	Morgan Stanley Reports		Equity Betas to 3/1/17		Financial Data on 12/31/16 (\$mil)		
	Report	WACC	1 Year	2 Years	Cash	Debt at	Equity at
	Date		Daily	Weekly		Book Value	Market Value
Snap Inc.	3/27/2018	9.7%					
Alphabet	3/23/2017	8.0%	0.99	1.34	\$12,918	\$3,935	\$539,070
Amazon	1/18/2017	7.5%	0.97	1.30	\$19,334	\$20,413	\$356,313
eBay	1/19/2017	6.3%	1.31	1.38	\$1,816	\$8,960	\$33,191
Etsy	3/1/2017	8.1%	1.57	2.32	\$182	\$12	\$1,361
Facebook	2/2/2017	8.6%	0.86	1.12	\$8,903	\$0	\$331,594
Groupon	2/16/2017	8.2%	1.95	2.08	\$863	\$228	\$1,896
GrubHub	2/8/2017	8.5%	1.48	1.13	\$240	\$0	\$3,220
LinkedIn (a)	4/29/2016	9.1%	n/a	n/a	n/a	n/a	n/a
Priceline Group	2/28/2017	8.0%	1.45	1.33	\$2,081	\$7,169	\$72,343
Twitter	2/9/2017	6.3%	0.91	1.71	\$989	\$1,687	\$11,563
Yelp	11/3/2016	8.3%	1.63	1.46	\$272	\$0	\$2,992
Zynga	1/19/2017	9.0%	1.18	1.22	\$852	\$0	\$2,292
Average		8.0%	1.30	1.49			
Median		8.2%	1.31	1.34			

Source: Individual equity research reports for each firm by Morgan Stanley, available on ThompsonOne, accessed 3/30/18. The betas and financial data are from Standard & Poor's Capital IQ database, accessed 4/6/18.

Note (a): Because Microsoft acquired LinkedIn in late 2016, financial and trading data was not available.

Exhibit 12 Summary of Morgan Stanley Investment Ratings, March 2017

Rating Category	Coverage Universe		Coverage of		IB Clients as % of All Ratings
	(All Ratings)		Investment Banking (IB)		
	Count	Percent	Count	Percent	
Overweight/Buy	1,148	35%	286	43%	25%
Equal-weight/Hold	1,418	43%	297	45%	21%
Not-Rated	61	2%	8	1%	13%
Underweight/Sell	638	20%	76	11%	12%
Total	3,265	100%	667	100%	

Source: Nowak, B., et al., "Crackle or Pop? Initiate OW, \$28 PT," Snap Inc. analyst report p. 38, Morgan Stanley Research, 3/27/17.

Appendix List of Equity Analyst Reports on Snap Inc. Used in this Case Study

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- Anthony, V., "Rating: Hold. Formally Establishing a Hold Rating," Snap Inc. Analyst Report, **Aegis Capital Group**, 3/2/2017.
- Blackledge, J., et al., "Initiate SNAP at Outperform," Snap Inc. Analyst Report, **Cowen and Company**, 3/27/2017.
- Boone, R.V., & Josey, R.V., "Initiating Coverage at Market Outperform Rating & \$28 Target: Benefiting from Media Fragmentation as Engagement Grows," Snap Inc. Analyst Report, **JMP**, 3/27/2017.
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Walmsley, L., et al., "Not Afraid of Ghosts; Initiate with Buy Rating and \$30 TP," Snap Inc. Analyst Report, **Deutsche Bank**, 3/27/2017.

Wieser, B., "Snap Inc.: Can't Get there from here. Initiating Coverage with SELL Rating and \$10 Price Target," Snap Inc. Analyst Report, **Pivotal**, 3/2/2017.

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